SafeInvest: Suitability, Cost-Aware Rebalancing, and the Limits of Automation

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Team Lebmar, No. 87 17 October 2026

Executive summary

SafeInvest must make suitable, explainable investment decisions without pretending that automation can guarantee a customer's loss tolerance. We build a transparent three-asset catalogue, an empirical suitability screen, and a sequential policy that trades only when estimated net benefit exceeds an declared conservatism margin. A distinct review state pauses trading while preserving all subsequent investment outcomes in the evaluation.

Our frozen targets, in equity/bond/bill order, are Conservative 15/5/80, Balanced 45/10/45, and Growth 90/5/5 percent. They are selected on development data under declared risk preferences, not chosen to force the five customers into expected labels. The frozen onboarding rule refers A and D, assigns B and E to Conservative, and assigns C to Balanced. The last assignment is borderline and is not validated as a reliable automatic recommendation: its loss-exceedance rate rises from 12/120 in development to 10/60 in the final scenarios. C therefore warrants human confirmation before unattended deployment; we retain, rather than retrospectively rewrite, the frozen rule and its measured outcomes.

In the 60 final 30-year scenarios, the selected fixed-target, cost-gated policy with review has median real annual growth of 3.66%, compared with 3.38% for annual equal-weight rebalancing. It trades in 35.5% of reviews and incurs 46.2% less mean cumulative fee expenditure. However, its 90th-percentile maximum real drawdown is 43.44%, versus 39.52% for the benchmark. This is a growth–risk trade-off, not comprehensive superiority.

Bounded automatic target updates lower some drawdown estimates but reduce median growth. Their incremental utility interval includes zero in both architecture validation and final evaluation. We therefore keep the frozen fixed-target policy and do not recommend automatic target updates on this evidence. Review is a governance safeguard, not an effective stop-loss: 48.3% of final paths enter review, and losses can continue.

What is established. Financial identities and causal timing pass automated tests; all outcomes remain in the sample, including review periods; source, data and configuration hashes tie final results to a frozen model. Cost sensitivity preserves the main qualitative trade-off.

What is not established. Neither real-world suitability, a guaranteed loss ceiling, reliable market prediction, nor universal policy dominance is established by 180 synthetic annual scenarios. We treat the failure to validate C's borderline assignment as an important result of a trustworthy system.

1 Accounting and purchasing-power risk

Each scenario begins with 100 units in bills. There are no subsequent deposits or withdrawals, short sales, leverage, taxes, other charges or terminal liquidation. At the beginning of year t, a decision sees only that scenario's observations through $t-1$. Returns for year t are applied after any trade and fee. All three assets are observed jointly with inflation.

Let x_i be pre-trade nominal asset values and $V = \sum_i x_i$. For a target $w_i \geq 0$, $\sum_i w_i = 1$, we explicitly adopt a fee-after allocation convention. Money entering destination assets is charged once; sale proceeds also fund the fee. Thus, with $c = 0.0025$,

$$C = c \sum_i [(V - C)w_i - x_i]_+, \qquad x_i' = (V - C)w_i. \tag{1}$$

This scalar, piecewise-linear equation is solved exactly over active destination assets. It ensures $\sum_i x_i' = V - C$. Transferring an actual 20 units costs 0.05. A 20% *fee-after target* is slightly different because the investable wealth has fallen. Keeping current holdings costs zero; initial allocation uses the same equation. This payment convention is a declared implementation choice where the problem does not specify the funding asset.

With observed nominal returns $r_{i,t}$,

$$V_t = \sum_i x_{i,t}'(1 + r_{i,t}), \quad I_t = \prod_{u=1}^{t}(1 + \pi_u), \quad R_t = V_t / I_t, \quad R_0 = 100. \tag{2}$$

Define annual net real returns $z_t = R_t / R_{t-1} - 1$. For horizon H we report

$$G_H = (R_H/100)^{1/H} - 1, \qquad \sigma_H = \mathrm{sd}(z_1, \ldots, z_H), \tag{3}$$

$$D_H = \max_{0 \leq t \leq H} \left(1 - \frac{R_t}{\max_{0 \leq u \leq t} R_u}\right). \tag{4}$$

Growth is actual purchasing-power growth after fees. Standard deviation describes ordinary fluctuations; maximum drawdown describes a large peak-to-trough loss, with larger values meaning greater loss. Including R_0 prevents omission of first-year allocation costs. Relative real drawdown is our explicit definition; we do not import a Brownian market assumption from the drawdown literature [6].

All drawdowns use annual observations. A fee reduces the subsequent net wealth, but we do not add a fee-only intrayear observation to the drawdown series. This keeps forecasting and reporting consistent. Actual intrayear drawdowns may be larger and cannot be recovered from the supplied data.

For each scenario, metrics are calculated first and then summarized across scenarios. The P90 of D_H is therefore the 90th percentile of scenario-level maximum drawdowns, not a percentile of annual returns. The bill proxy is not real-risk-free: inflation can reduce its purchasing power. This prevents equating a high bill allocation with guaranteed preservation.

2 A product catalogue and a suitability process

We consider 5-percentage-point simplex weights, require at least 5% in each asset, and add the exact equal-weight target. This gives 172 candidates. Fixed targets are annually restored during catalogue fitting, including all costs. The catalogue uses the declared objective

$$J_H(w;\lambda) = \mathbb{E}[\log(R_H(w)/100)] - \lambda \mathbb{E}[D_H(w)]. \tag{5}$$

At $H = 30$, coefficients $\lambda = 10, 4, 1$ express increasingly growth-oriented preferences. They are policy assumptions, not customer parameters identified from three questionnaire fields. Bill floors are 30%, 10%, and 5%, respectively. The original 1–80 fit and final 1–120 refit use the same rule. The Balanced refit changes from 60/15/25 to 45/10/45, illustrating estimation sensitivity; the neighbourhood in the fitting grid is comparatively flat.

Product	Equity	Bond	Bill	λ
Conservative	15%	5%	80%	10
Balanced	45%	10%	45%	4
Growth	90%	5%	5%	1

Conservative and Growth restore their fixed targets annually. Balanced uses the selected cost-aware policy. The names describe relative product risk, not guaranteed loss ranges or compulsory assignments.

For customer j, we use exactly $H_j = 5, 15, 30, 25, 6$ years for A–E. A product passes our empirical loss screen if

$$\widehat{p}_j = \frac{1}{120} \sum_{s=1}^{120} \mathbf{1}\{D_{s,H_j} > L_j\} \leq 0.10. \tag{6}$$

Bill floors for high/medium/low liquidity are 30/10/0 percent. Among passing products, choose the highest development median real CAGR; if none pass, refer. These are transparent service policies, not organizer-imposed thresholds or 90% safety guarantees. No withdrawal sufficiency is tested because the experiment has no withdrawals. Wilson intervals accompany exceedance rates.

Client	Frozen action	H	Tolerance	Explanation
A	Human review	5	10%	No product passes loss screen
B	Conservative	15	25%	2/120 exceedances
C	Balanced	30	40%	12/120; exactly at cutoff
D	Human review	25	15%	Long horizon does not offset low tolerance
E	Conservative	6	35%	Both liquid choices pass; higher fitted growth

For A and D this is a statement about the available catalogue, not proof that every possible investment rule is unsuitable. In fitting-only checks, no fixed grid target met their strict screen under the stated liquidity floor. Changing the cutoff to 5% moves C to Conservative; at 20%, A becomes Conservative and B Balanced. These differences expose policy sensitivity.

3 Cost-aware annual decisions and human review

We use four unguarded strategies to separate effects and two guarded variants:

Rule	Definition
S0	Equal-weight target, restore annually (required baseline)
S1	Chosen Balanced target, restore annually
S2	Same fixed target, trade only after net-benefit test
S3	Same executor, allow bounded target changes
S2+R, S3+R	Add the same explicit review/pending rule

S1–S0 separates target choice; S2–S1 execution; S3–S2 target updates. Guarded versus corresponding unguarded results measure the effect of pending behaviour. This avoids attributing an allocation change to clever timing.

Forecasting assumption. From fitting scenarios only, sample 256 joint paths by concatenating six continuous five-year blocks. Assets and inflation stay together; random seed is 1729. At year t, use the remaining $H - t + 1$ years of these paths. All actions share identical forecast paths. This assumes development blocks remain representative and is not evidence of return predictability. Conditional market timing is deliberately omitted.

Action comparison. Current holdings are normalized by current wealth. For each possible action, apply its self-financing fee, project drifting holdings without additional forecast trades, and compute

$$Q_t(a) = \widehat{\mathbb{E}}[\log(R_H^a / R_{t-1})] - 4\widehat{\mathbb{E}}[D_H^a]. \tag{7}$$

The drawdown extends the actual historical peak and already realized maximum drawdown. The future cannot erase a past loss. The best action is taken only if $Q_t(a) > Q_t(\text{keep}) + 0.002$. The margin is in total utility units; cost is already reflected in wealth and is not subtracted twice. Initially, only bills versus the base catalogue target are compared; the dynamic neighbourhood opens from the second review. Rejecting all actions keeps both current holdings and the approved target unchanged.

For S3, equity targets are the original Balanced equity weight plus $\{-0.10, -0.05, 0, 0.05, 0.10\}$. Non-equity weights retain their original bond/bill ratio; bill weight must remain at least 10%. A single target change cannot exceed 5 percentage points. Global targets therefore cannot drift indefinitely. Actual holdings can drift beyond target bounds between trades.

Review and pending. In the generic Balanced experiment, an observed real drawdown of at least 25% triggers review at the next annual decision. Customer-specific diagnostics instead use that customer's stated tolerance. While review is pending, no further trading or target updates occur; returns and all scoring continue. Without human-decision data we keep the path pending through its remaining horizon, rather than inventing perfect human decisions. The rule does not cap losses at the threshold. Invalid inputs also require suspension; the input loader rejects missing or non-finite observations rather than treating them as zero returns.

4 Development, validation and a frozen decision

The organisers supply 180 independent synthetic scenarios, each with 30 annual periods, based on selected U.S. JST Release 6 series [1–3]. Raw historical database observations are not used. Scenarios 1–80 fit the catalogue and forecast bank; 81–120 compare fixed architectures. The fixed risk coefficients, fee rule, 0.002 utility margin, forecast specification and bounded action set are written in the protocol. No validation parameter search is performed.

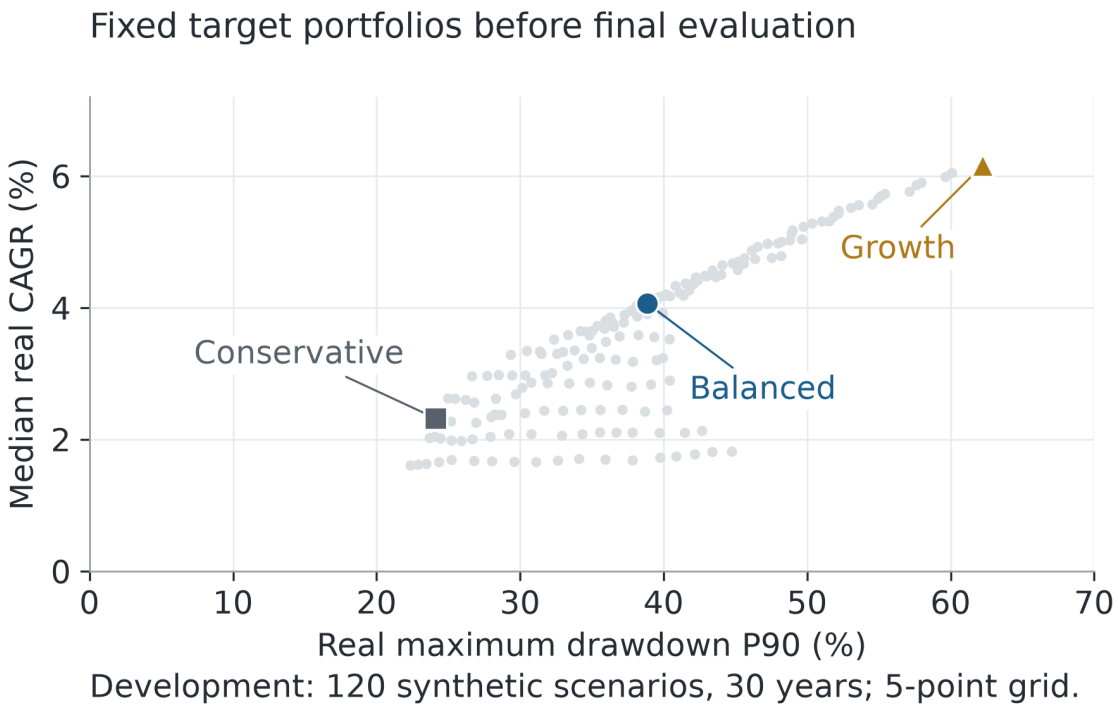


Figure 1: Development catalogue: growth versus large-loss risk, with the final refitted targets marked. Synthetic contest data, scenarios 1–120. These are fitting diagnostics, not out-of-sample performance.

The architecture rule enables S3+R only if its paired mean utility advantage over S2+R has a positive 95% bootstrap lower endpoint, its P90 drawdown rises by no more than 2 percentage points, and P10 CAGR falls by no more than 0.25 percentage points. The latter two are empirical screens, not statistical noninferiority guarantees. The 4,000 paired resamples preserve complete scenarios and use a fixed seed.

Validation yields incremental mean utility 0.0440 with interval $[-0.0063, 0.0962]$. S3+R therefore fails the first criterion. We select S2+R, refit weights and the forecast bank once on 1–120, and freeze the architecture, customer assignments, sources, protocol, data and forecast hashes. The final 60 paths are then evaluated without further adjustment. Equal-weight rules remain valuable comparators under estimation error [5]; this does not prove equal weighting optimal in our synthetic task. Cost-aware sequential decisions are motivated by [4], without importing a convexity or optimality claim.

Exposure and development corrections. Before formal development, a source inventory printed pooled descriptive statistics over all 180 paths. No strategy selection used them, but this is not a completely pristine blind test. Independent review corrected initial-allocation gating and a forecast versus annual-drawdown measurement mismatch; development was rerun before any final policy evaluation. The 1–120 suitability estimates are refitted, in-sample diagnostics. Final customer results audit them without rewriting their frozen assignments. These qualifications are retained in the audit log.

5 Final evaluation: gains come with trade-offs

Table 1 uses all 60 final paths, including the 100-unit starting point and every pending-review year. Percentages are net real results; trade rate includes initial reviews. S2+R is the frozen policy.

Table 1: Thirty-year final results, all entries in percent. G: real CAGR; D: maximum real drawdown.

Rule	Median G	P10 G	Median D	P90 D	Trade rate
S0	3.38	1.50	22.29	39.52	100.0
S1	3.68	1.62	27.44	44.23	100.0
S2	3.79	1.50	25.46	46.00	38.8
S3	3.49	1.67	21.68	41.44	43.2
S2+R	3.66	1.61	25.46	43.44	35.5
S3+R	3.40	1.54	21.68	40.11	39.2

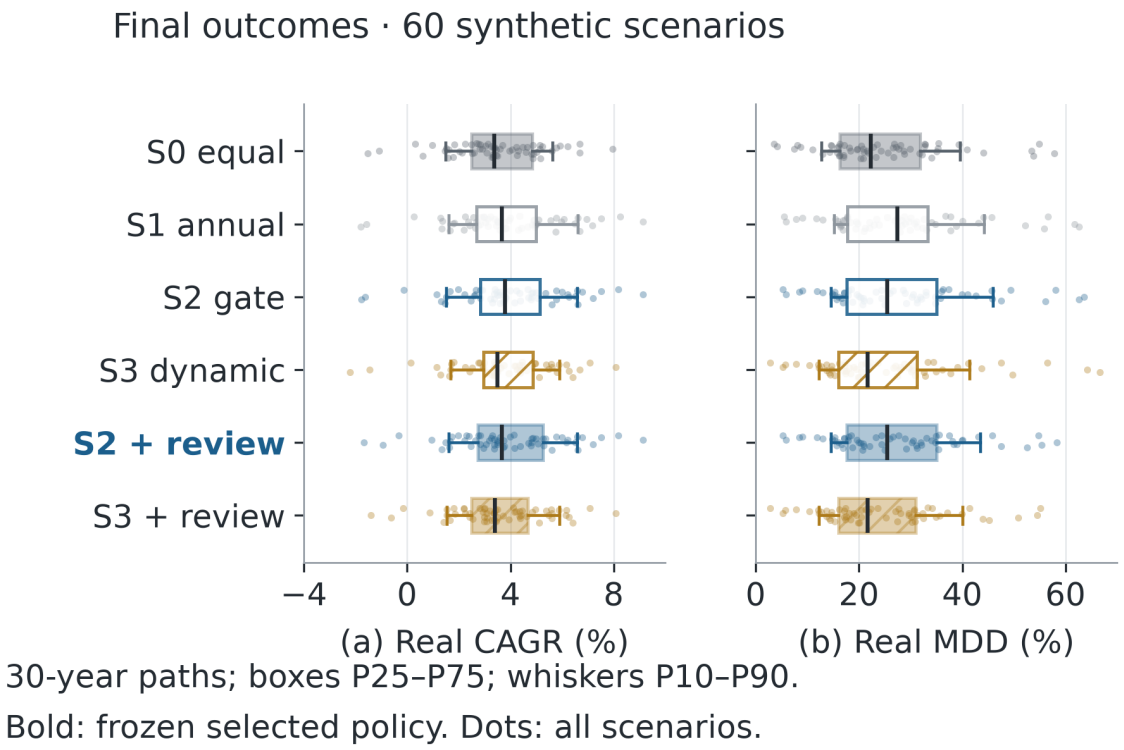


Figure 2: Final scenario distributions. Spread across scenarios is distinct from uncertainty in a mean or quantile; displayed distribution ranges are not confidence intervals. Synthetic contest data, $n = 60$.

S2+R's median growth difference from S0 is 0.282 percentage points, with paired interval $[0.026, 0.758]$. Its P90 drawdown difference is $+3.92$ points, with interval $[-2.18, 10.95]$. Mean utility difference is 0.0232 with interval $[-0.0530, 0.0971]$. Thus higher typical growth does not establish an unambiguous risk-adjusted improvement. Mean fees are 0.526 versus 0.979 nominal units; this is fee expenditure, not the full compounded opportunity cost. S2+R trades in 35.5% of all reviews, or 33.28% after excluding initial deployment.

S3+R versus S2+R has utility difference 0.0218, interval $[-0.0215, 0.0650]$: insufficient evidence to enable updates. Its median growth is lower by 0.260 points while P90 drawdown is lower by 3.33 points. We keep the validation decision; we do not switch architectures after seeing the final table.

6 A severe scenario and an honest safeguard

We preselect the final scenario with the largest S0 drawdown, rather than choose the path where our policy wins most. This is scenario 155.

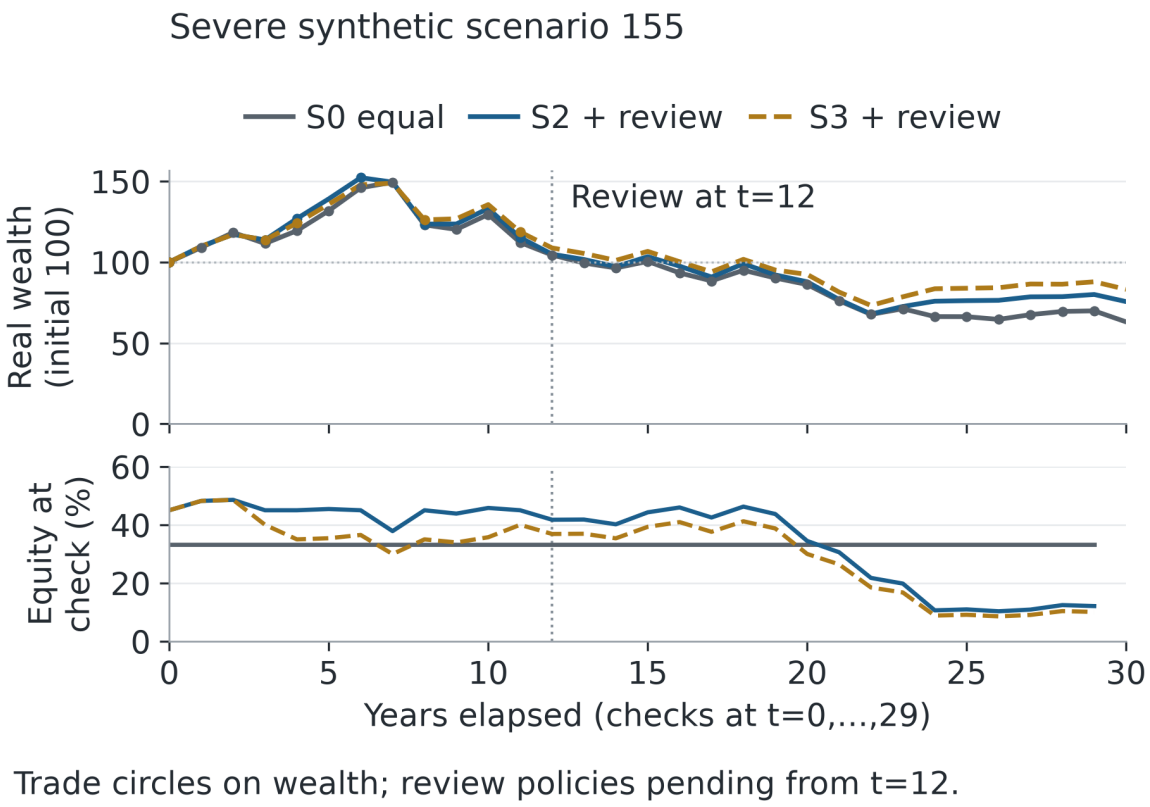


Figure 3: Scenario 155: actual purchasing-power wealth and equity exposure. Decisions for a year precede its returns. Review starts at period 13 after observing the first 12 years; pending is not a liquidation.

Rule	Real terminal wealth	Maximum drawdown	Trades
S0	63.00	57.81%	30
S2	58.27	62.58%	13
S2+R	75.55	55.42%	6
S3+R	83.11	50.83%	5

S2+R ends below its original purchasing power despite its higher terminal wealth than S0 on this path. Its worst drawdown occurs in year 22, after review begins. A pause therefore does not insure the customer's wealth. Across all paths, 29/60 enter S2+R review, raising a material human-capacity issue.

The review ablation lowers P90 drawdown from 46.00% (S2) to 43.44% (S2+R), but the paired difference interval is $[-6.75, +1.70]$ percentage points. Neither this nor its utility interval establishes a general loss-reduction effect. Its defensible role is halting unsupported automated decisions.

Customer-facing message (37 words):

Your portfolio's purchasing-power loss has reached our review threshold. We have paused automatic trading and requested an adviser review. Your current investments remain exposed to market changes; this pause does not guarantee recovery or prevent further losses.

7 Reliability, customer calibration and limitations

We vary transaction costs without refitting weights, forecasts or decision settings. The chosen rule remains S2+R; only its actual costs and consequently permitted decisions change.

Table 2: Frozen-rule cost sensitivity, 60 final 30-year scenarios; all entries other than rule labels are percentages.

Fee rate	Rule	Median G	P90 D	Trade rate
0.125	S0	3.383	39.50	100.0
0.125	S2+R	3.664	43.44	35.8
0.250	S0	3.376	39.52	100.0
0.250	S2+R	3.658	43.44	35.5
0.500	S0	3.362	39.55	100.0
0.500	S2+R	3.647	43.44	35.1

The growth advantage and higher tail drawdown versus equal weights persist over this range. This supports the same qualified conclusion, not universal robustness. For a separately recomputed 15-year horizon, S2+R median CAGR is 3.730%, versus 3.763% for S0; P90 drawdown is 33.99% versus 34.10%. The 30-year growth advantage cannot be generalized to every customer horizon.

Table 3: Frozen customer assignments audited at their own horizons. These exceedance rates are empirical, not loss guarantees.

Client	Frozen product	Development	Final	Final upper 95%
B	Conservative, 15 years	2/120	4/60 (6.7%)	15.9%
C	Balanced, 30 years	12/120	10/60 (16.7%)	28.0%
E	Conservative, 6 years	0/120	0/60 (0.0%)	6.0%

The upper endpoints are from two-sided 95% Wilson intervals. C's final point estimate no longer passes the 10% screen, although its interval still includes 10%; this is not statistical proof of a true exceedance rate above 10%. It is insufficient evidence for reliable unattended suitability. We preserve the frozen C assignment to expose this weakness and recommend human confirmation before deploying it. C's diagnostic review trigger is its own 40% tolerance, not the generic experiment's 25%, so its P90 drawdown (46.00%) is not the main-table 43.44%.

Important limitations are annual observation, synthetic representativeness, small tail samples, policy-dependent risk preferences and liquidity floors, and a stationary block forecast. Overlapping blocks do not create additional independent evaluation scenarios. Real customers' income, reserves and withdrawal timing are absent. Pending behaviour is mechanical; no evidence about an adviser's actual intervention benefit is available.

8 Recommendation and reproducibility

Use the 15/5/80 Conservative, 45/10/45 Balanced and 90/5/5 Growth targets as an explicit catalogue for this study. Refer A and D; assign B and E to Conservative. The frozen rule assigns C to Balanced, but the borderline development screen does not retain its pointwise calibration in final data: require human confirmation before unattended use. Growth need not be assigned to any of these five profiles. Questionnaire tolerance is not a promise to cap losses.

For Balanced, retain fixed targets with cost-aware execution and an explicit review state. Do not enable automatic target updates on the current evidence. The proposed rule trades less and has higher 30-year median growth than the required benchmark, but its tail drawdown is higher and the utility advantage is uncertain. The main safeguard is suspension of unsupported automatic decisions with clear customer communication; holding during review can still lose money and needs practical human-response capacity.

The supplementary archive contains source code, input attribution, fixed protocol, model fingerprints, per-scenario paths, all reported tables, figure builders, report source and an independent result audit. Financial identities, future-information perturbations, target bounds and review-state tests are automated. Reproduction stages are `develop`, `freeze` and `final`; existing frozen/final archives cannot be silently overwritten. The public repository is `https://github.com/Lebmar/GKG-MCM`.

References

[1] HKUST(GZ) MCM organisers (2026). *Senior Group Problem A: SafeInvest*, supplied statement and synthetic dataset. CC BY-NC-SA 4.0.

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Use of AI Tools

OpenAI Codex assisted with problem interpretation, model design, Python code, independent AI review, experimental analysis and English drafting. Automated tests and independent numerical checks were used to verify accounting, causal timing and reported calculations; bibliographic records were checked against primary sources. These checks do not replace the author's review. The author remains responsible for correctness, attribution and the submission.